PAC-Former — mandatory PAC-interaction tokenizer + tri-axial encoder

TUEV supervised, seed 0: $\kappa = 0.5493$ vs. raw-token backbone 0.4679 (AGENT.md §13.43-I); both 1,635,734 parameters

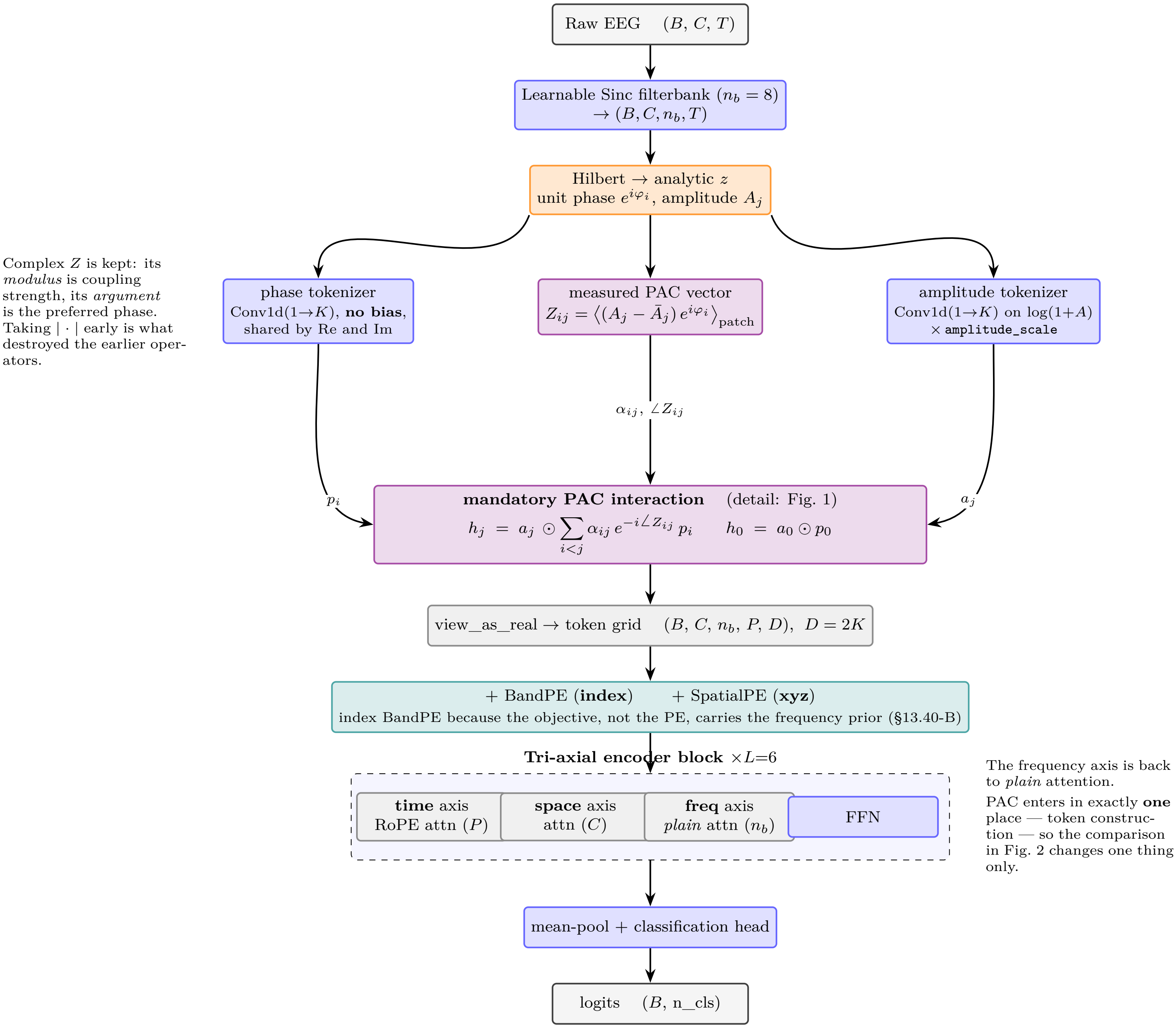

Fig. 1 — the mandatory, phase-reference-invariant PAC token

for each electrode c and 1 s patch p, over n_b bands: unit phase $e^{i\varphi_i}$, amplitude A_j

$$Z_{ij} \;=\; \frac{1}{L}\sum_{t\in p}\big(A_j(t)-\bar{A}_j\big)\,e^{i\varphi_i(t)} \qquad \alpha_{ij} = \frac{|Z_{ij}|}{\sum_{i'<j}|Z_{i'j}|} \qquad \text{valid iff } i<j$$

amplitude is patch-centred (dPAC debiasing); `valid = tril(diagonal=-1)` restricts every edge to *slow phase → fast amplitude* — the MI mixers of §13.41 scored all n_b^2 pairs instead, a contamination of the treatment definition.

$$p_i = \mathcal{L}(\cos\varphi_i) + i\,\mathcal{L}(\sin\varphi_i)$$

one shared **bias-free** real conv $\mathcal{L}$

$$a_j = \texttt{scale} \odot \mathcal{A}(\log(1+A_j))$$

real conv $\mathcal{A}$, its own bias

$$\boxed{h_j \;=\; \underbrace{a_j}_{\text{target amplitude}} \odot \sum_{i<j} \alpha_{ij}\; \underbrace{e^{-i\angle Z_{ij}}}_{\text{preferred-phase alignment}}\; \underbrace{p_i}_{\text{slower-band phase}}}$$

$h_0 = a_0 \odot p_0$ (the slowest band has no slower driver: it is the *root* of the directed hierarchy, and the only gauge-covariant token)

Why the prior cannot be routed around. There is no raw high-band token beside h_j, no `pac_scale`, no gate, no PAC branch, no auxiliary loss. Every sample-dependent feature reaching the encoder for $j > 0$ contains *both* the target amplitude and an aligned slower-band phase. BandPE/SpatialPE are constants and cannot carry sample-specific EEG around the product. This is the §13.18 lesson applied structurally: a prior with a free path beside it gets optimised away, so we removed the path.

Phase-reference (gauge) invariance.
Shift band i's phase origin by an arbitrary δ_i:

$$\mathcal{L} \text{ is real-linear and bias-free} \Rightarrow \quad p_i \;\mapsto\; e^{i\delta_i} p_i,$$
$$Z_{ij} \;\mapsto\; e^{i\delta_i} Z_{ij} \;\Rightarrow\; e^{-i\angle Z_{ij}} \;\mapsto\; e^{-i\delta_i} e^{-i\angle Z_{ij}}$$

the two rotations cancel exactly, so $h_{j>0}$ is **unchanged**. Verified numerically: $\max|\Delta| = 8.3 \times 10^{-7}$ over bands 1..7 (`scratchpad/verify_pacint_configs.py`).

This is a property of the operator, not something training must discover — and it is the concrete difference from the old `FreqPhaseSteered` mixer, which rotated learned real feature pairs that had no defined phase reference.

Fig. 2 — the three parameter-matched arms (TUEV, seed 0, 1,635,734 params each)

`measured` $\alpha_{ij} = \frac{|Z_{ij}|}{\sum |Z_{\cdot j}|}$, phase factor $e^{-i\angle Z_{ij}}$ $\kappa = \mathbf{0.5493}$

`scramble` same $|Z_{ij}|$, but *which edge owns which preferred phase* is freshly permuted every forward $\kappa = 0.5099$

`uniform` $\alpha_{ij} = 1/|\{i < j\}|$, no phase factor — the product topology survives, the measurement does not $\kappa = 0.4734$

raw-token baseline per-(c, b) Conv1d on the filtered waveform; no interaction at all $\kappa = 0.4679$

+0.0394
preferred-phase ownership

+0.0365
measured PAC magnitude

+0.0055 — **tie**
the product topology alone is worth nothing

Reading. The decisive control is `uniform`, not `measured`. It keeps the entire mandatory amplitude × slower-phase structure and only swaps the measured coefficients for a uniform average — and it *ties* the raw baseline. So the gain is not the multiplicative token structure; it is the measured quantity poured into it. The two contrasts above then split that gain into two nearly equal, separately attributable halves.

Not yet established. `measured`−`scramble` $= +0.0394$ is the only evidence that *preferred phase specifically* matters; it is barely 2× the 0.02 single-seed noise line, and `scramble` redraws its permutation every forward while `measured` does not (val std 0.0466 vs 0.0382), so part of it may be injected stochasticity. Needs seeds $1, 2$ and a *deterministic* phase-destruction arm (real α_{ij}, no $e^{-i\angle Z_{ij}}$) before the preferred-phase claim is written.